DATA MINING CAPSTONE PROJECT

Classifying Philippine Household Personal Care & Lifestyle Spending Tiers

Using Machine Learning



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Problem Statement & Research Objectives

Despite personal care, cosmetics, and toiletries representing a major market segment in the Philippines, the specific socio-economic and demographic factors that distinguish high-spending households from low-spending households remain underexplored. Necessitating a supervised binary classification approach using Philippine Statistics Authority (PSA) Family Income and Expenditure Survey (FIES) data to accurately predict household personal care spending tiers.

CORE OBJECTIVES

- ① Engineer balanced binary Spending_Tier target
- ② Compare 4 ML classifiers across 2 tools
- ③ Identify top socio-economic spending drivers

TARGET STAKEHOLDERS

- **FMCG & Retail Marketers**
  Segment targeting & regional campaigns
- **Consumer Economists**
  Discretionary spend across income strata
- **Policy Researchers**
  Social protection & welfare monitoring



Dataset & Cloud Data Engineering (BigQuery)

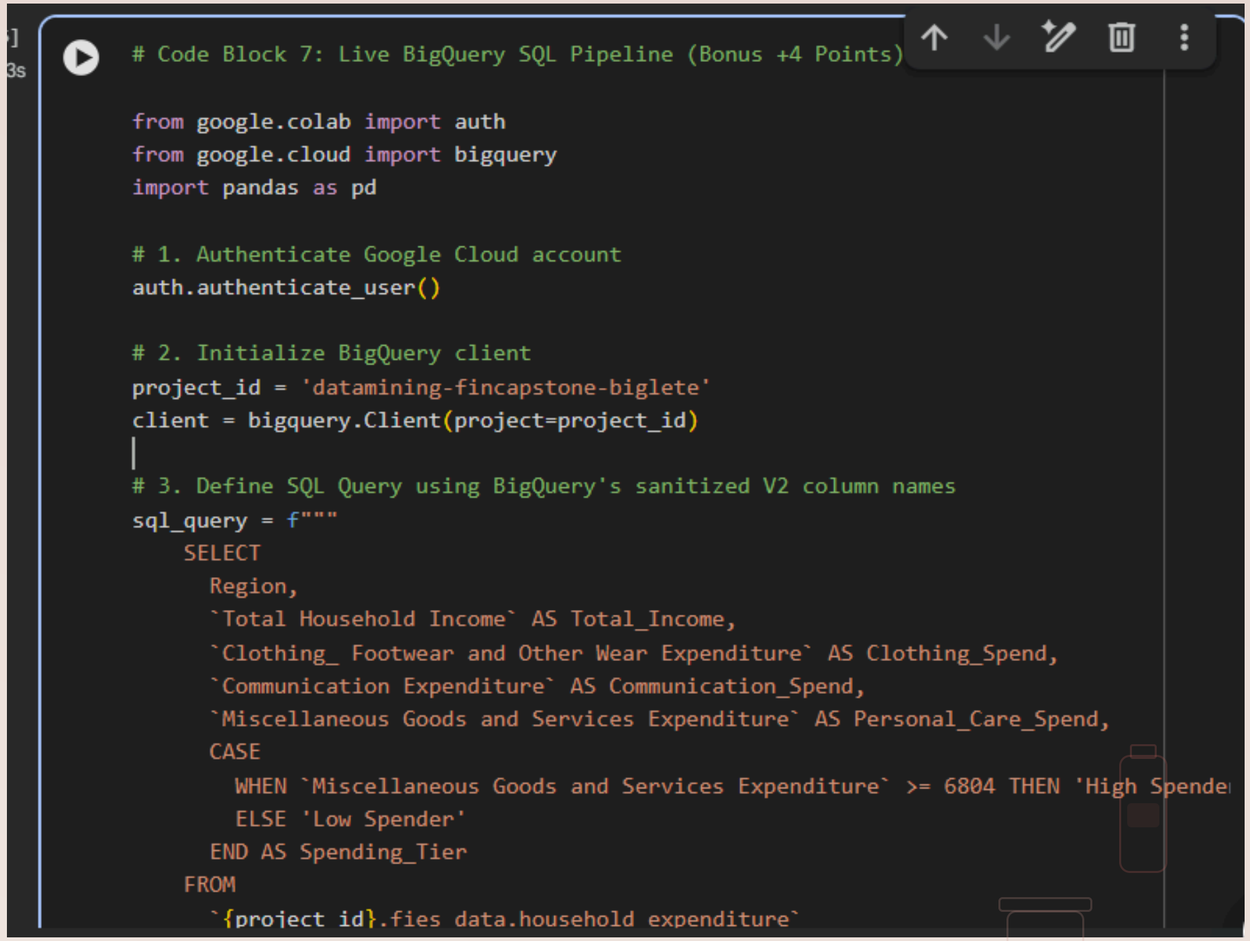
PSA FIES DATASET

Records	41,544 household observations
Original Features	60 attributes
Regions	17 Philippine administrative regions
Missing Values	0 — dataset is complete
Target Column	Miscellaneous Goods & Services Expenditure
Source	PSA / Kaggle (Grosven, 2018)

BIGQUERY SQL — MEDIAN-SPLIT CLASSIFICATION (₱6,804)

```
CASE
  WHEN Misc_Goods_Expenditure >= 6804
  THEN 'High Spender'
  ELSE 'Low Spender'
END AS Spending_Tier
```

BIGQUERY PIPELINE SCREENSHOT



```
# Code Block 7: Live BigQuery SQL Pipeline (Bonus +4 Points)

from google.colab import auth
from google.cloud import bigquery
import pandas as pd

# 1. Authenticate Google Cloud account
auth.authenticate_user()

# 2. Initialize BigQuery client
project_id = 'datamining-fincapstone-biglete'
client = bigquery.Client(project=project_id)

# 3. Define SQL Query using BigQuery's sanitized V2 column names
sql_query = f"""
    SELECT
      Region,
      `Total Household Income` AS Total_Income,
      `Clothing_ Footwear and Other Wear Expenditure` AS Clothing_Spend,
      `Communication Expenditure` AS Communication_Spend,
      `Miscellaneous Goods and Services Expenditure` AS Personal_Care_Spend,
      CASE
        WHEN `Miscellaneous Goods and Services Expenditure` >= 6804 THEN 'High Spende
        ELSE 'Low Spender'
      END AS Spending_Tier
    FROM
      `{project_id}.fies_data.household_expenditure`
```

Target Engineering & Exploratory Data Analysis

SPENDING_TIER BINARY TARGET — MEDIAN SPLIT AT ₱6,804

50.04%
High Spender ≥ ₱6,804

49.96%
Low Spender < ₱6,804

Near-perfect class balance — no imbalance bias in training

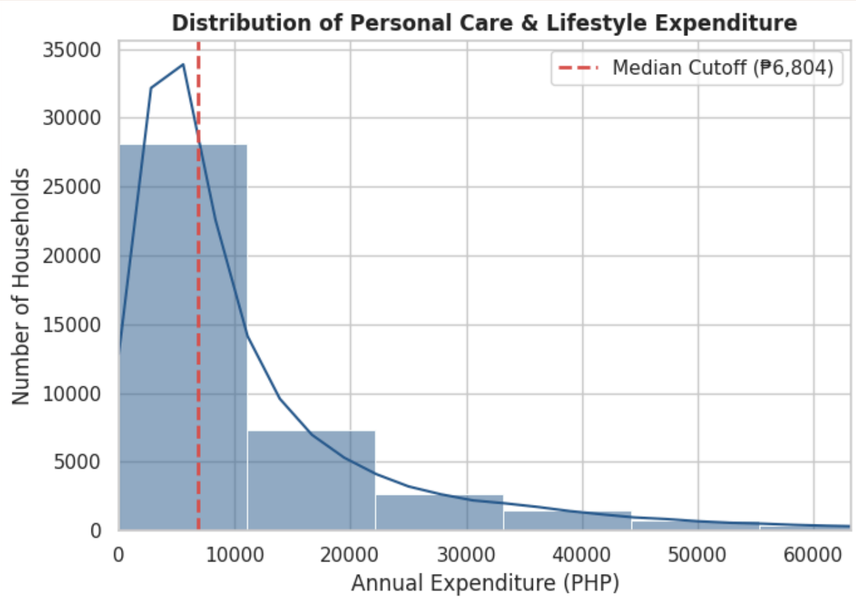
15 SELECTED SOCIO-ECONOMIC FEATURES

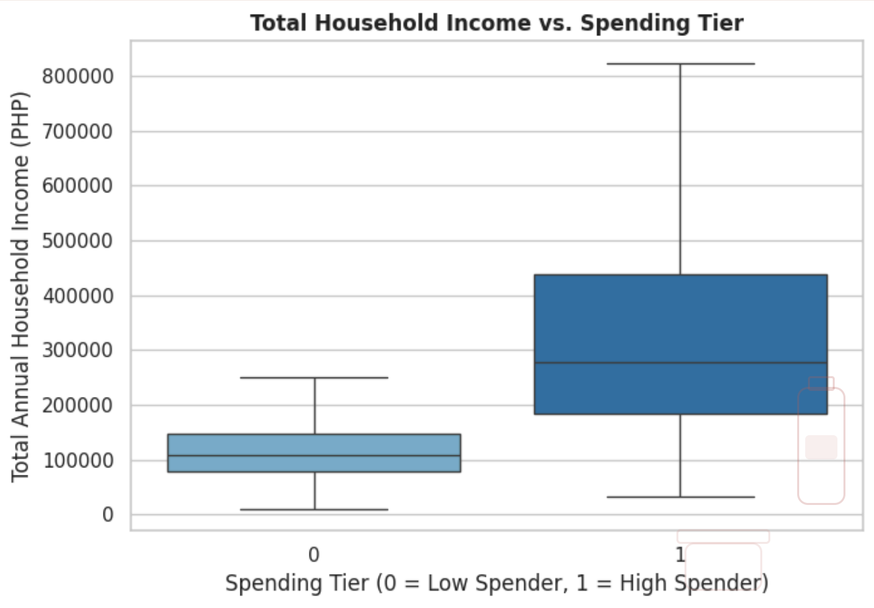
- Region (17 PH regions → one-hot encoded)
- Total Household Income
- Clothing, Footwear & Other Wear Expenditure
- Communication · Housing · Food · Transport …
- Household Head Age · Family Size · Employment

PREPROCESSING PIPELINE

- One-hot encode Region → 30 final columns
- StandardScaler (Z-score) on all continuous features
- Zero missing values — no imputation required

EDA VISUALISATIONS





Dual-Tool Data Mining Methodology

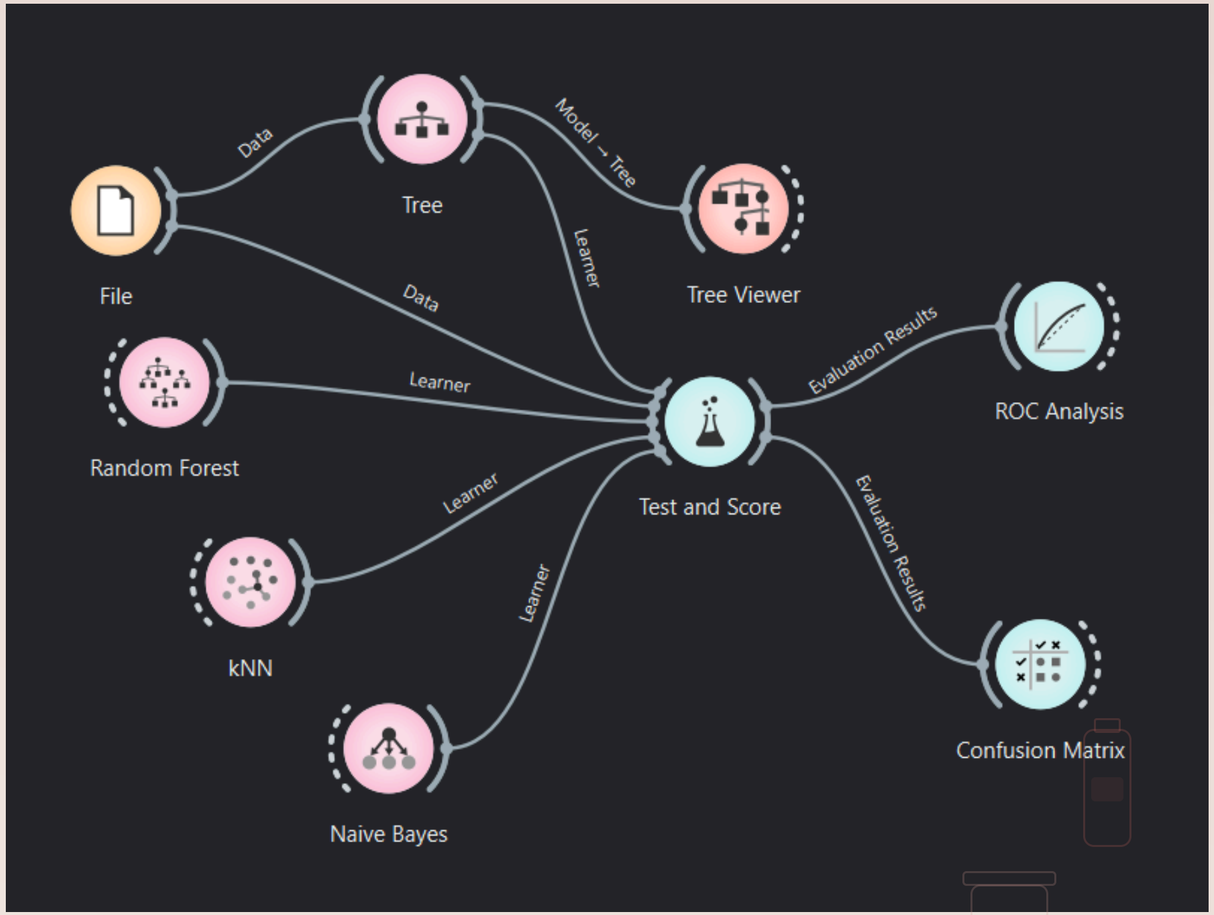
FOUR CLASSIFICATION ALGORITHMS

Random Forest	Ensemble · 100 trees · max_depth=12
Decision Tree	Rule-based · max_depth=10
k-NN	Instance-based · k=7 · Euclidean
Naïve Bayes	Probabilistic · GaussianNB

DUAL VALIDATION SETUP

	Python scikit-learn	Orange Data Mining
Split	80 / 20 stratified	10-fold stratified CV
Test records	8,309 held-out	Full 41,544 (CV)
Metrics	Accuracy · Precision · F1	AUC · CA · MCC

ORANGE DATA MINING WORKFLOW



Model Performance Comparison

PYTHON SCIKIT-LEARN — 80/20 STRATIFIED HOLD-OUT

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	84.31%	84.96%	83.41%	84.17% ★
Decision Tree	81.27%	81.36%	81.17%	81.27%
k-NN	81.07%	84.75%	75.81%	80.03%
Naïve Bayes	77.05%	88.96%	61.81%	72.94%

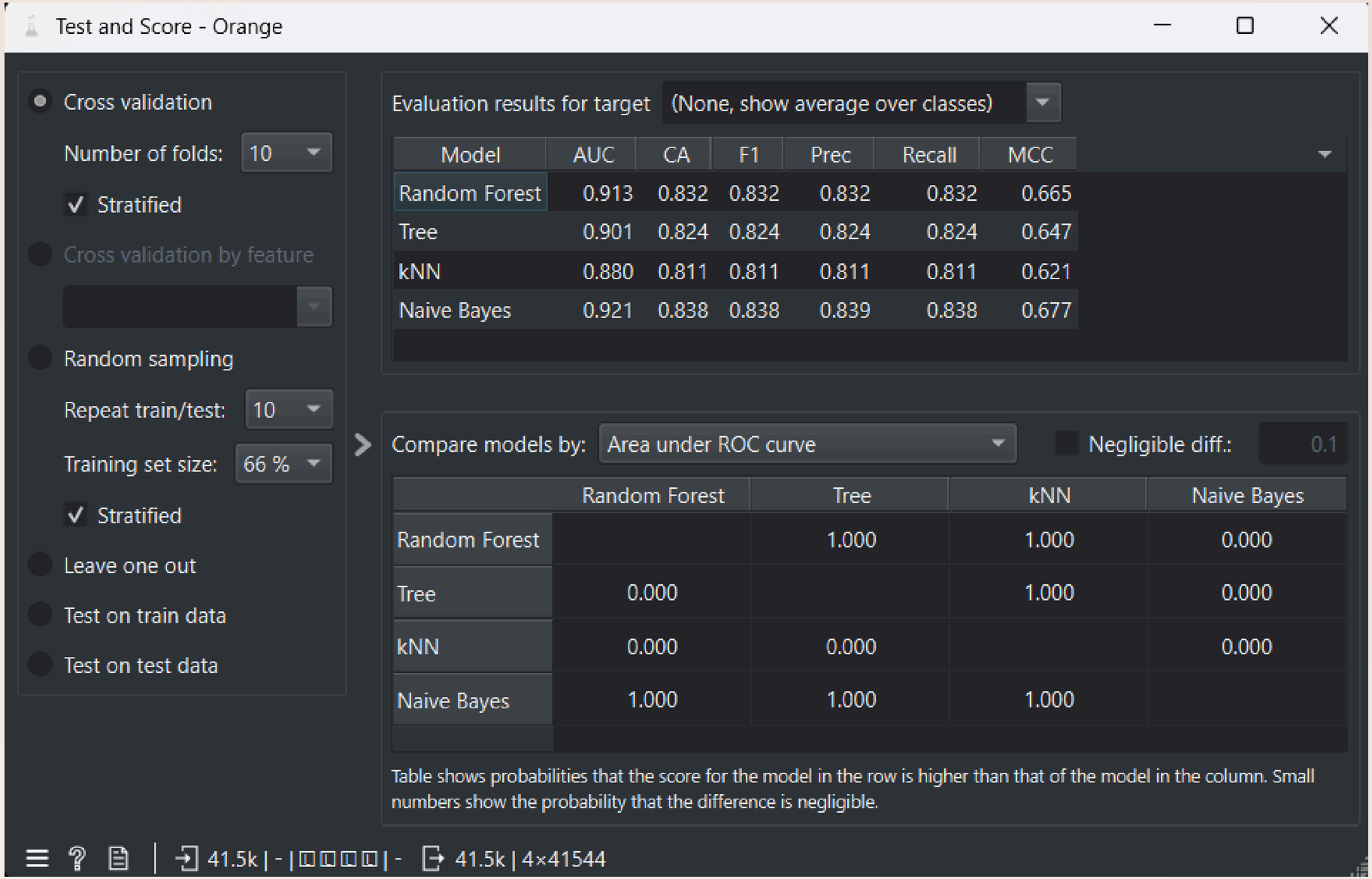
ORANGE DATA MINING — 10-FOLD STRATIFIED CV

Model	AUC	CA	F1	MCC
Naïve Bayes	0.921★	83.8%	0.838	0.677
Random Forest	0.913	83.2%	0.832	0.665
Tree	0.901	82.4%	0.824	0.647
kNN	0.880	81.1%	0.811	0.621

Key Takeaway

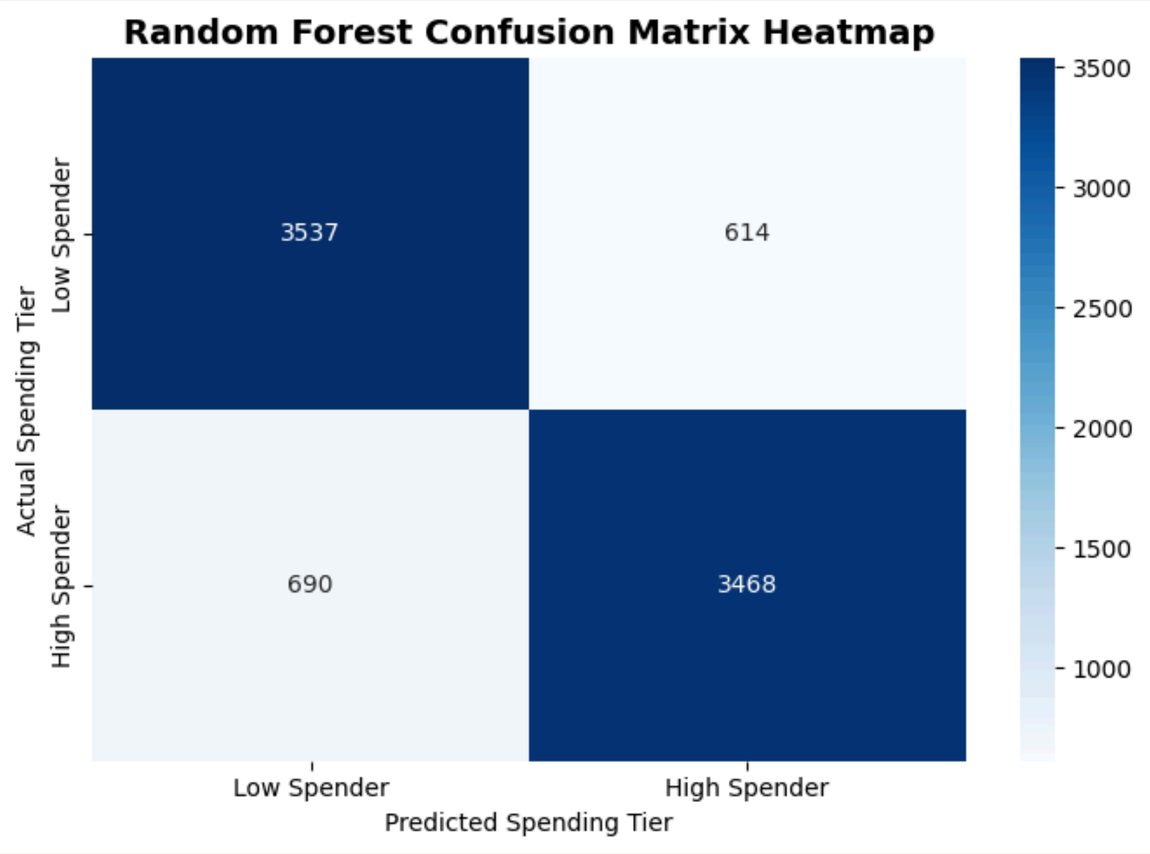
Random Forest delivers the most balanced performance across both tools — 84.3% accuracy (Python) · 83.2% (Orange CV) · AUC 0.913 · balanced precision & recall. Naïve Bayes leads on AUC (0.921) in CV but underperforms on Recall (61.8%) in hold-out testing. → Random Forest is the recommended production model.

ORANGE TEST & SCORE WIDGET OUTPUT

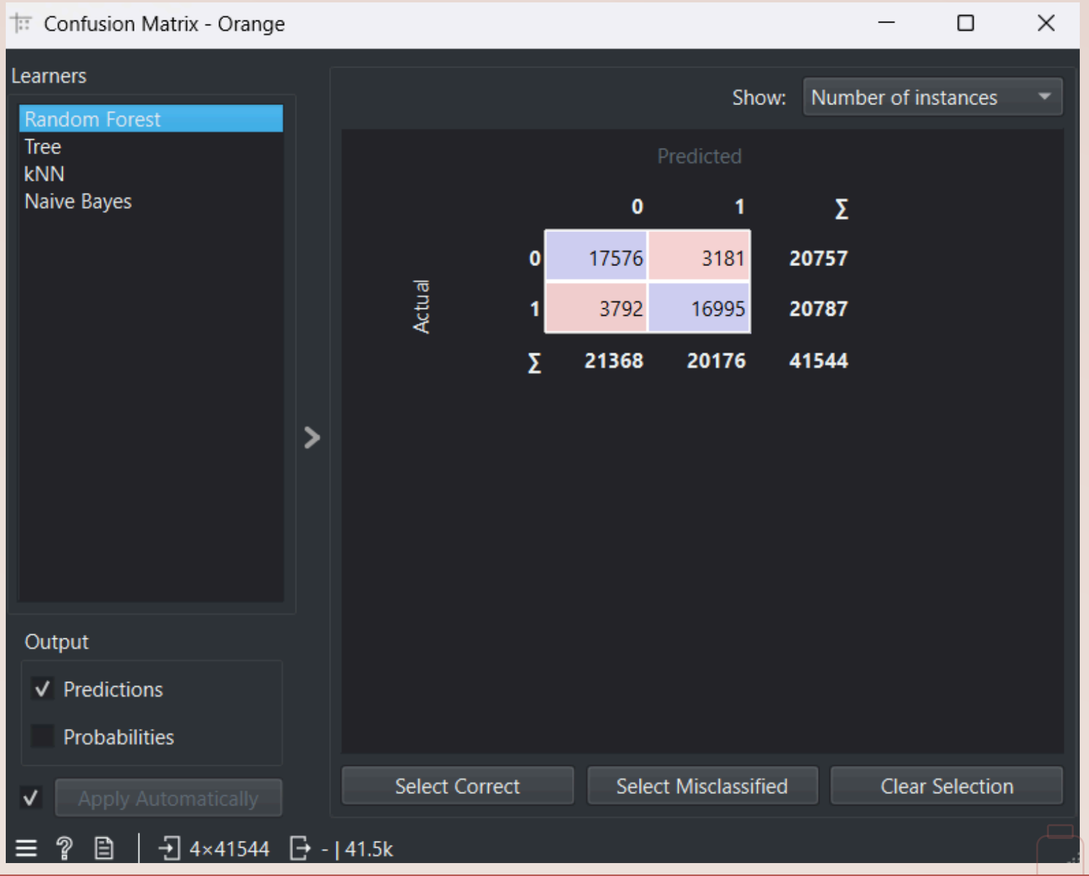


Classification Diagnostics: Confusion Matrices

PYTHON SCIKIT-LEARN · TEST SET (8,309 RECORDS) — ACC 84.3%



ORANGE · 10-FOLD CV (41,544 RECORDS) — ACC 83.2%



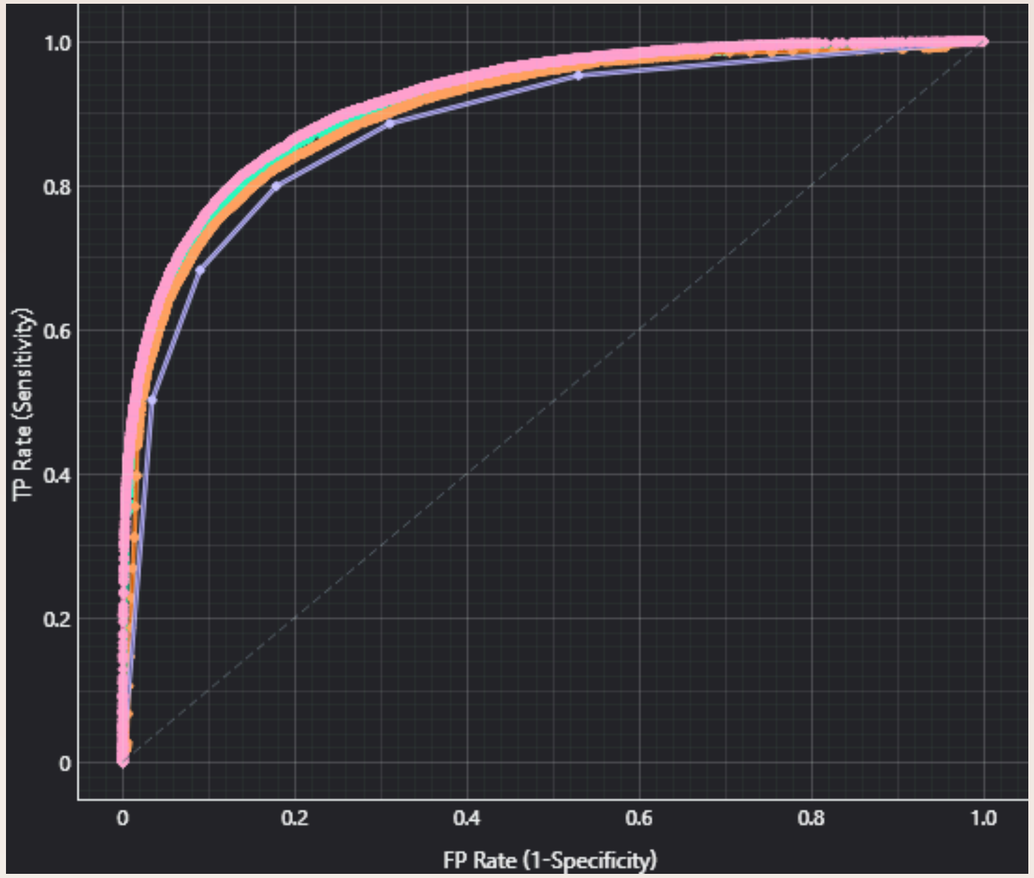
① Consistent accuracy across both evaluation frameworks (84.3% vs 83.2%)

② Balanced errors — FP=614 vs FN=690 (Python) · no systematic directional bias

③ Low False Negatives → minimal missed High Spenders = strong FMCG targeting precision

ROC Discrimination & Decision Tree Logic

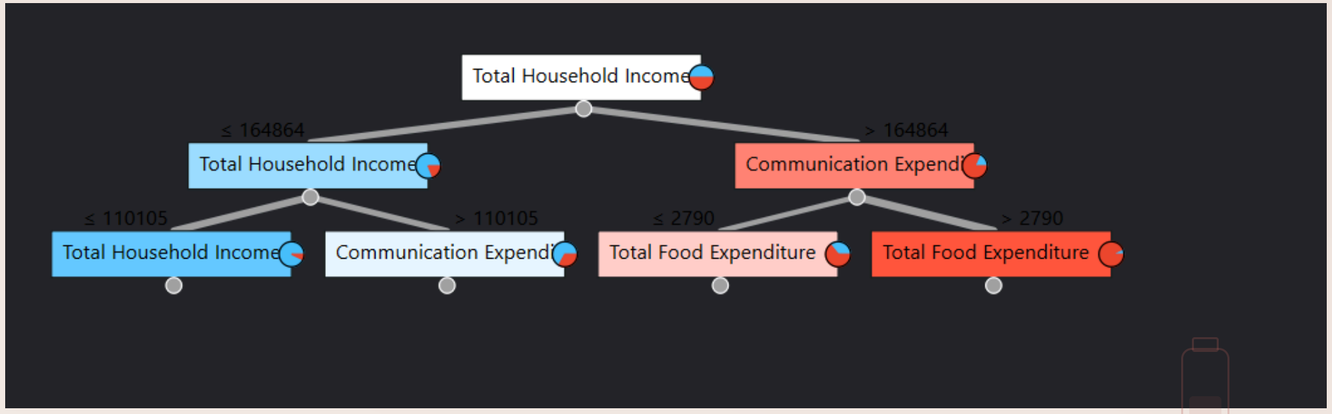
ORANGE ROC ANALYSIS — ALL 4 MODELS



Model	AUC
Naïve Bayes	0.921
Random Forest	0.913
Tree	0.901
kNN	0.880

All models well above chance baseline (AUC 0.5)

ORANGE DECISION TREE VIEWER (MAX_DEPTH=8)



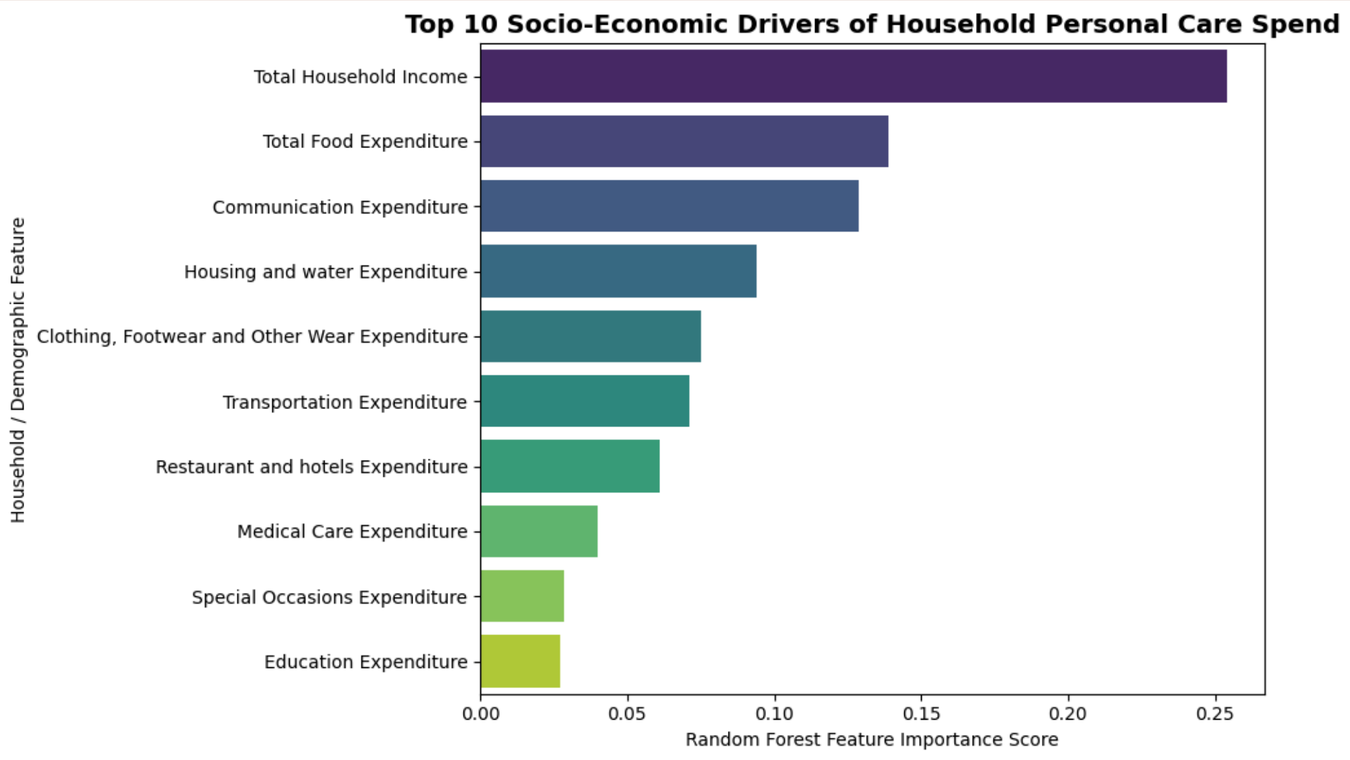
→ **Root split: Total Household Income ≤ / > ₱164,864**

→ 2nd-level: Communication Expenditure ≤ / > ₱2,790

→ 3rd-level: Total Food Expenditure confirms High Spender

Top Socio-Economic Drivers of Spending Tier

RANDOM FOREST FEATURE IMPORTANCE (TOP 10)



TOP 5 PREDICTORS & CONSUMER INSIGHTS

① Total Household Income ~0.26

Primary gatekeeper — income > ₱164K drives High tier

② Total Food Expenditure ~0.14

Reflects overall household consumption scale

③ Communication Expenditure ~0.13

Urban connectivity proxy → digital-lifestyle signal

④ Housing & Water Expenditure ~0.10

Urban residence → access to personal care retail

⑤ Clothing & Footwear Expenditure ~0.08

Direct co-predictor of personal care investment

Strategic Recommendations & Limitations

ACTIONABLE RECOMMENDATIONS

- Target households earning > ₱164,864/yr for premium personal care campaigns
- Use Communication Expenditure as a digital-first marketing signal
- High housing-spend regions = urban market proxies for retail expansion
- Segment on Income + Clothing spend for SKU-level product planning
- Policy: monitor Misc. Goods spend as a household welfare indicator

KEY METHODOLOGICAL LIMITATIONS

Cross-sectional data

Single snapshot — no year-on-year dynamics

Self-reported survey

Recall bias & income underreporting likely

Category proxy

Miscellaneous ≠ pure personal care category

Regional granularity

Masks urban/rural variation within regions

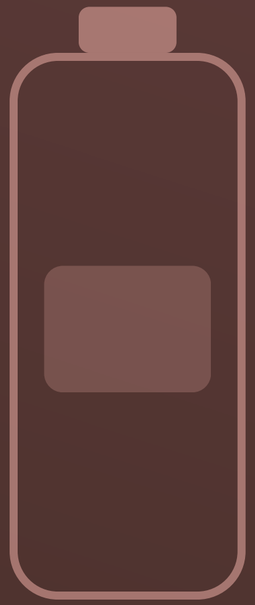
Temporal validity

2018 Kaggle release — post-pandemic shifts not captured



CONCLUSION & REFLECTION

Key Takeaways



① Random Forest is the strongest model

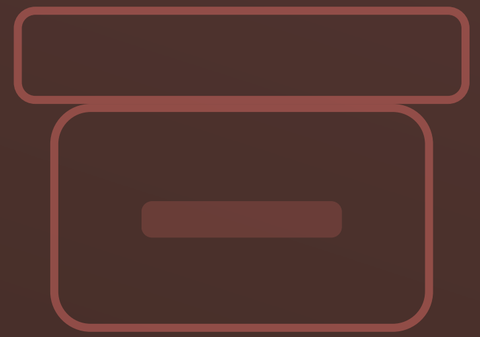
84.3% (Python) · 83.2% (Orange CV) · AUC 0.913 — balanced across all metrics

② Total Household Income is the #1 driver

Households earning > ₱164K/yr are significantly more likely to be High Spenders

③ Dual-tool validation adds scientific rigour

Python hold-out & Orange 10-fold CV converge — results are robust, not overfitted



This project demonstrates an end-to-end cloud-to-ML pipeline: BigQuery ingestion → Python scikit-learn classification → Orange visual cross-validation — bridging data engineering and consumer analytics for FMCG & policy decision-making.

Thank You!

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